

Rayna Yu

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EDUCATION

Northeastern University

Boston, MA

B.S in Computer Science and Mathematics | PlusOne: M.S in Computer Science | Minor: Robotics

April 2028

GPA: 4.0/4.0 | Honors: Dean's List, Dean's Scholarship

Relevant Coursework: Algorithms (Graduate-level), Systems, Theory of Computation, Artificial Intelligence, Matrix Methods in Data Analysis and Machine Learning, Linear Algebra, Probability and Statistics, Object-oriented design

SKILLS

Languages: Java, Python, TypeScript, JavaScript, GraphQL, SQL, MATLAB, HTML/CSS, Racket

Frameworks: React, NestJS, Next.js, FastAPI, LangChain, NumPy, Scikit-learn, Pandas, Matplotlib, PyTorch

Tools & Technologies: Git/GitHub, Jira, AWS, Visual Studio Code, Datadog, Playwright, Cypress, IntelliJ, Eclipse IDE, Linux

EXPERIENCE

Liberty Mutual Solaria Labs

January 2026 - June 2026

Software Development Co-op

Boston, MA

- Contributed to Liberty Mutual's Certainly platform EFT payment pilot across **5** microservices, coordinating and deploying a simultaneous zero-downtime production launch that resulted in a **46.3%** EFT adoption rate within the first few weeks, directly supporting initiatives to reduce over **\$100M** in annual card processing fees
- Built a production monitoring triage assistant using **LangChain**, designed to automatically parse **Datadog** alerts, classify affected services, retrieve supporting context from **Datadog**, **Jira**, and **GitHub** via **REST APIs**, and return structured JSON triage recommendations, reducing manual first-pass investigation burden
- Designed and implemented an automated DLQ redrive system using **AWS Lambda** and **SQS**, replacing a manual CLI-based recovery process by scheduling automatic message retries with **S3** quarantine, reducing unpredictable interventions

Code 4 Community

January 2025 - Present

Software Developer

Boston, MA

- Engineered full-stack applications for nonprofits (GI Boston, Securing Safe Food, NEFAC, etc.), converting requirements into functional sites using **React**, **Next.js**, and **PostgreSQL**, improving data accessibility and operational efficiency
- Collaborated in a 7-person Agile team to streamline backend operations for a recruitment portal, implementing database optimizations with **pgAdmin 4**, and contributing through sprint planning, **GitHub** workflows, and peer code reviews

Teuscher Lab

May 2025 - August 2025

Undergraduate Researcher

Remote

- Analyzed **over 38,000** Boston Vision Zero crash records and additional large-scale datasets to develop machine learning models predicting vehicle-pedestrian crashes using **Scikit-learn**, achieving **94% accuracy** with a Random Forest model
- Integrated the trained model into a mobile navigation system using multi-source **GeoJSON** datasets and **OpenRouteService API** to generate context-aware routes, resulting in improved pedestrian safety and accessible routes by accounting for environmental hazards beyond speed and distance

Northeastern University; Computer Science Department

May 2025 - December 2025

Teaching Assistant, CS 2800: Logic and Computation and CS 1800: Discrete Structures

Boston, MA

- Supported 100+ students in CS 2800 and CS 1800, teaching complex concepts such as property-based testing, code specification, set theory, data structures, and discrete mathematics using **Python** and **OCaml**
- Led weekly recitations and office hours, mentoring students in formal reasoning, proofs, and computational logic, targeting common areas of difficulty and resulting in noticeable improvement on homework, quizzes, and exams

PROJECT EXPERIENCE

AI Pet Agent - Python, NLP, PyTorch, Hugging Face

August 2025 - Present

Machine Learning Engineer

- Developed a modular GenAI agent for pet owners that interprets natural language queries, retrieves relevant information from large document sets using TF-IDF, and generates structured responses via a Hugging Face language model.

Horizon Lab and NUAerospace - C++, C, Embedded Systems

August 2025 - December 2025

Flight Software Developer

- Contributed to hands-on engineering development projects for Arduino-based flight software in C/C++ to monitor rocket internal states in real time, ensuring reliable operation and enabling precise fault detection and localization during launches